

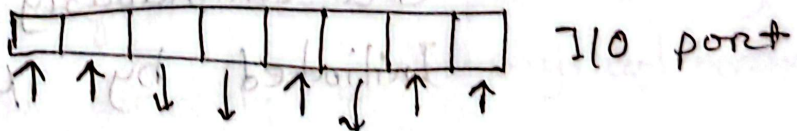
9.3: program I/O

A μp communicates with an external device via one or more registers called I/O ports using programmed I/O. I/O port usually of two types.

For one type, each bit in the port can be set up as an input or output, normally by writing a 0 or a 1 in the corresponding bit of the data-direction register.

If an 8-bit data-direction register contains 39H, the corresponding port is defined as

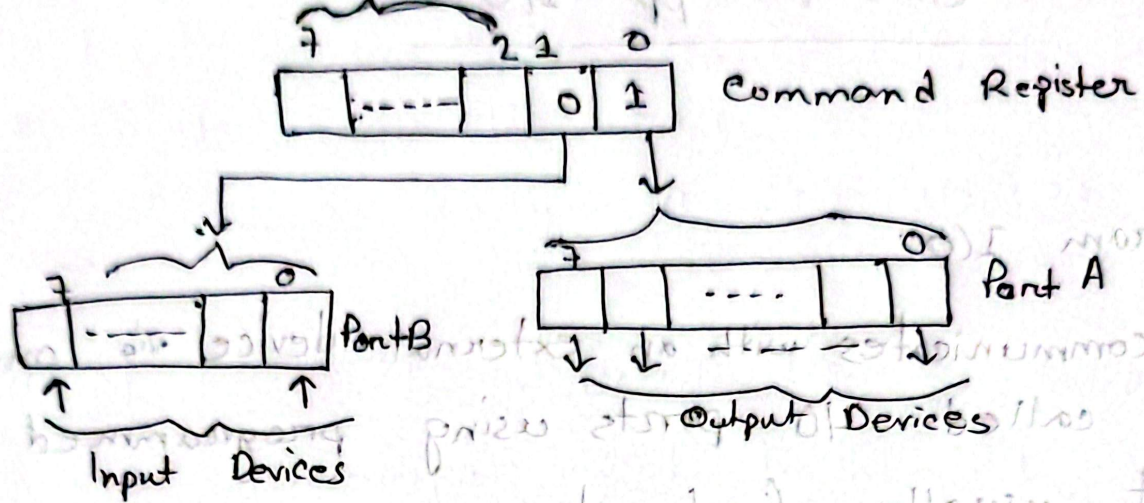
7	6	5	4	3	2	1	0	Bit position
0	0	1	1	0	1	0	0	Data-Direction Reg.



The port contains the actual input or output data. The data direction register is an output register and can be used to configure the bits in the port as input or outputs.

for parallel I/O, there is only one data-direction register for all ports. A particular bit in the data direction register configures all bits in the port as either input or outputs.

Other control information such as timer control signals



Ex 1.1 Three types of I/O:

① Program I/O: CPU-controlled I/O method where processor manage data transfer.

CPU continuously checks device status.
Initiated by CPU.

② Interrupt-Driven: I/O method where device interrupts CPU when ready. CPU services device only when interrupt occurs. Initiated by device.

③ Direct-Memory Access: I/O method where data transfer directly between device and memory. CPU not involved during transfer. Initiated by device.

4.2 Basic difference between standard I/O and memory mapped I/O:
Standard I/O: I/O devices are accessed using a separate address space from memory.
Instruction: IN, OUT

Memory-mapped I/O: I/O devices are treated as memory locations.
Instruction: MOV, LOAD, STORE

What are PIR (Peripheral Interrupt Request) and DMAC (Direct Memory Access)?

4.3 Programmed I/O: Program I/O is a technique where the μp itself controls all data transfers between I/O devices and memory by executing instructions.

Virtual I/O: Virtual I/O is a technique where the user interacts with logical devices provided by the OS, instead of directly with physical hardware.

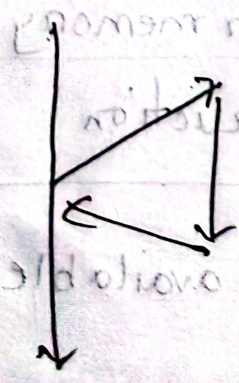
4.4 Feature	Memory Mapping	Memory-Mapped I/O
Definition	Assigning addresses to RAM/ROM	Assigning memory address to I/O devices.
Purpose	Used for memory storage	Used for communication with device
Address Space	Separate for memory	Shared with I/O
Instruction	Memory instruction	Same memory instruction
Effect	Full memory available	Reduce available memory

4.5 Polled I/O: In polled I/O, the μp repeatedly check the status of an I/O device to determine wheathor it is ready for data transfer.

Interrupt I/O: In interrupt I/O, the device sends an interrupt signal to the CPU when it is ready and the CPU responds by executing an interrupt service routine (ISR).

4.6 Subroutine: A subroutine is a program block that is called explicitly by the main program using a call instruction.

4.7 Interrupt Address vector: An interrupt address vector is the memory location that stores the starting address of the interrupt service Routine, (ISR).



9.8

Feature	Maskable Interrupt	Non-Maskable Interrupt
Definition	Interrupt that can be enable or disable by software.	Interrupt that <u>cannot</u> be disable.
Control	can be masked	cannot be masked
Priority	Lower	Highest
usage	Normal I/O operation	Critical situation
Ex:	Keyboard Interrupt	Power failure.

9.10

Feature	Internal Interrupt	External Interrupt
Definition	Interrupt generated inside CPU due to program execution.	Interrupt generate by external hardware device
Source	Inside CPU	Outside CPU
Cause	Program error	Hardware signals
Ex:	Arithmetic error	Keyboard, I/O device.

Q: If we use 2-byte pixel value in a 24-bit lookup table representation, how many byte does the lookup table occupy?

Sol:

Each lookup table entry

$$24 \text{ bits} = 3 \text{ bytes}$$

If pixel values are 2 bytes

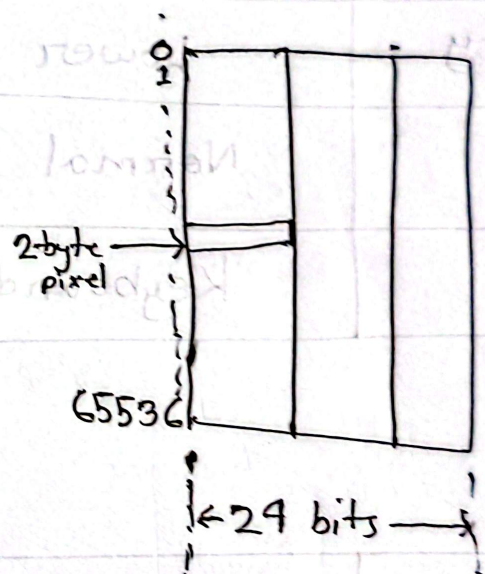
each pixel represent

$$2^{16} = 65536$$

So, total lookup table size,

$$65536 \times 3 \text{ bytes}$$

$$= 196608 \text{ bytes}$$



Lookup table